serviceRegistryManagement

Service Description

Service ID: *"serviceRegistryManagement"*

Abstract
This document describes the `serviceRegistryManagement` service, version 1.0.0, produced by the ServiceRegistry core system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `serviceRegistryManagement` service, which provides operator management of the whole ServiceRegistry content: devices, systems, service definitions, service instances and interface templates. It is produced by the ServiceRegistry system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

Corresponds to the Arrowhead 4.x ServiceRegistry `mgmt` endpoints.

2.2 How This Service Is Meant to Be Used

Operators and authorized management systems (e.g. the onboarding procedure, LocalCloudManager, Dynamic ServiceOrchestration) query and edit registry records on behalf of other systems, and maintain the interface templates that service registrations are validated against.

2.3 Important Delimitations and Dependencies

Access is governed by the management policy of the ServiceRegistry (`management.policy`: `SYSOP_ONLY`, `WHITELIST` or `AUTHORIZATION`); in the Arrowhead-test-cloud-java deployment it is a whitelist. Removing a device or system cascades to the records that depend on it as described per operation.

3 Service Interface

This section describes the interfaces to the `serviceRegistryManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

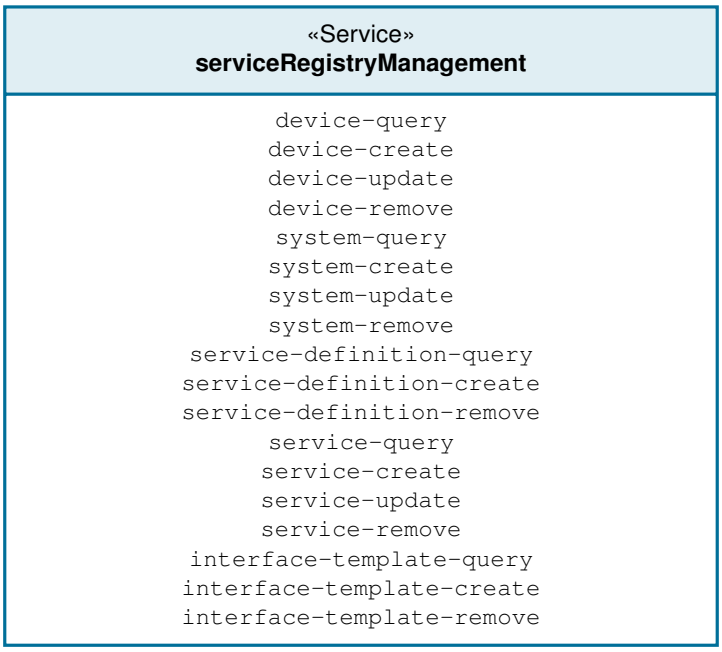


Figure 1: SysML-style block of the `serviceRegistryManagement` service and its operations.

The following interface operations are available.

3.1 operation device-query (DeviceQueryRequest) : DeviceListResponse

Returns a page of devices matching the filter.

3.2 operation device-create (DeviceListRequest) : DeviceListResponse

Creates devices.

3.3 operation device-update (DeviceListRequest) : DeviceListResponse

Updates the metadata and addresses of devices.

3.4 operation device-remove (List<Name>)

Removes devices by name.

3.5 operation system-query (SystemQueryRequest) : SystemListResponse

Returns a page of systems matching the filter.

Optional parameter: `verbose`.

3.6 operation system-create (SystemListRequest) : SystemListResponse

Creates systems on behalf of others.

3.7 operation system-update (SystemListRequest) : SystemListResponse

Updates systems.

3.8 operation system-remove (List<Name>)

Removes systems by name, with their service instances.

3.9 operation service-definition-query (Page) : ServiceDefinitionListResponse

Returns a page of service definitions.

3.10 operation service-definition-create (ServiceDefinitionListRequest) : ServiceDefinitionListResponse

Creates service definitions.

3.11 operation service-definition-remove (List<Name>)

Removes service definitions by name, with their service instances.

3.12 operation service-query (ServiceInstanceQueryRequest) : ServiceInstanceListResponse

Returns a page of service instances matching the filter, regardless of the discovery policy.
Optional parameter: `verbose`.

3.13 operation service-create (ServiceInstanceCreateListRequest) : ServiceInstanceListResponse

Creates service instances on behalf of providers. The reservation of core and support service definitions applies.

3.14 operation service-update (ServiceInstanceUpdateListRequest) : ServiceInstanceListResponse

Updates the expiry, metadata and interfaces of service instances.

3.15 operation service-remove (List<String>)

Removes service instances by identifier.

3.16 operation interface-template-query (ServiceInterfaceTemplateQueryRequest) : ServiceInterfaceTemplateListResponse

Returns a page of interface templates.

3.17 operation interface-template-create (ServiceInterfaceTemplateListRequest) : ServiceInterfaceTemplateListResponse

Creates interface templates with their property rules.

3.18 operation interface-template-remove (List<Name>)

Removes interface templates by name.

4 Information Model

Here, all data objects that can be part of the `serviceRegistryManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.32, which are used to represent things like hashes and identifiers.

4.1 struct Address

A network address with its type.

Field	Type	Description
type	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
address	String	One network address (host name, IPv4, IPv6 or MAC address).

4.2 struct DeviceListRequest

Devices to create or update.

Field	Type	Description
devices	List<DeviceRequest>	Devices to create or update.

4.3 struct DeviceListResponse

List of devices.

Field	Type	Description
entries	List<DeviceResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.4 struct DeviceQueryRequest

Paged filter for devices.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
deviceNames	List<Name>	Device names to match.
addresses	List<String>	Network addresses of the entity.
addressType	String	Address type filter: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
metadataRequirement List	List<MetadataRequirement>	Metadata requirements; each map is one alternative.

4.5 struct DeviceRequest

A device and its addresses.

Field	Type	Description
name	Name	Device name, PascalCase.
metadata	Metadata	Free-form metadata (JSON object).
addresses	List<String>	Network addresses of the entity.

4.6 struct DeviceResponse

A registered device.

Field	Type	Description
name	Name	Device name.
metadata	Metadata	Free-form metadata (JSON object).
addresses	List<Address>	Network addresses with their types.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.7 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

4.8 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

4.9 struct ServiceDefinitionListRequest

Service definitions to create.

Field	Type	Description
serviceDefinitionNames	List<Name>	Service definition names.

4.10 struct ServiceDefinitionListResponse

List of service definitions.

Field	Type	Description
entries	List<ServiceDefinitionResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.11 struct ServiceDefinitionResponse

A service definition.

Field	Type	Description
name	Name	Service definition name.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.12 struct ServiceInstanceCreateListRequest

Service instances to create on behalf of providers.

Field	Type	Description
instances	List<ServiceInstanceRequest>	Service instances to create or update.

4.13 struct ServiceInstanceInterfaceRequest

One interface of a service instance.

Field	Type	Description
templateName	Name	Name of the interface template, snake_case (e.g. `generic_https`).
protocol	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
policy	String	Security policy of the interface (`NONE`, `CERT_AUTH`, token policies).
properties	Map<String, Object>	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.14 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Field	Type	Description
templateName	Name	Name of the interface template, snake_case (e.g. `generic_https`).
protocol	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
policy	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
properties	Map<String, Object>	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.15 struct ServiceInstanceListResponse

List of service instances.

Field	Type	Description
entries	List<ServiceInstanceResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.16 struct ServiceInstanceQueryRequest

Paged filter for service instances.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
instanceIds	List<Identifier>	Instance identifiers to match.
providerNames	List<Name>	Provider system names to match.
serviceDefinitionNames	List<Name>	Service definition names.
versions	List<Version>	Service versions to match.
alivesAt	DateTime	Only instances that are still registered (not expired) at this UTC time are matched.
metadataRequirements List	List<MetadataRequirement>	Requirements on the service instance metadata; each map is one alternative.
addressTypes	List<String>	Address types the interfaces must offer (`HOSTNAME`, `IPV4`, `IPV6`, `MAC`).
interfaceTemplateNames	List<String>	Accepted interface templates, e.g. `generic_https`.
interfaceProperty RequirementsList	List<MetadataRequirement>	Requirements on the interface properties; each map is one alternative.
policies	List<String>	Interface security policies to match, e.g. `CERT_AUTH`.

4.17 struct ServiceInstanceRequest

A service instance and its provider.

Field	Type	Description
systemName	Name	Name of the provider system.
serviceDefinitionName	Name	Service definition name, camelCase (e.g. `temperatureReading`).
version	Version	Version of the service instance (default `1.0.0`).
expiresAt	DateTime	UTC expiry time; no value means no expiry.
metadata	Metadata	Free-form metadata (JSON object).
interfaces	List<ServiceInstanceInterface Request>	Service interfaces offered by the instance.

4.18 struct ServiceInstanceResponse

A registered service instance.

Field	Type	Description
instanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
provider	SystemResponse	The provider system.
serviceDefinition	ServiceDefinitionResponse	The service definition.
version	Version	Version of the service instance.
expiresAt	DateTime	UTC expiry time; no value means no expiry.
metadata	Metadata	Free-form metadata (JSON object).
interfaces	List<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.19 struct ServiceInstanceUpdateListRequest

Service instances to update.

Field	Type	Description
instances	List<ServiceInstanceUpdate Request>	Service instances to create or update.

4.20 struct ServiceInstanceUpdateRequest

Changes to one service instance.

Field	Type	Description
instanceId	Identifier	Instance to update.
expiresAt	DateTime	UTC expiry time; no value means no expiry.
metadata	Metadata	Free-form metadata (JSON object).
interfaces	List<ServiceInstanceInterface Request>	Service interfaces offered by the instance.

4.21 struct ServiceInterfaceTemplateListRequest

Interface templates to create.

Field	Type	Description
interfaceTemplates	List<ServiceInterfaceTemplate Request>	Interface templates to create.

4.22 struct ServiceInterfaceTemplateListResponse

List of interface templates.

Field	Type	Description
entries	List<ServiceInterfaceTemplate Response>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.23 struct ServiceInterfaceTemplateProperty

Rule for one interface property.

Field	Type	Description
name	Name	Name of the interface property.
mandatory	Boolean	Whether the property must be present in every interface of this template.
validator	String	Validator of the value, e.g. `PORT`, `NOT_EMPTY_ADDRESS_LIST`, `HTTP_OPERATIONS`.
validatorParams	List<String>	Parameters of the validator.

4.24 struct ServiceInterfaceTemplateQueryRequest

Paged filter for interface templates.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
templateNames	List<Name>	Interface template names to match.
protocols	List<String>	Protocols to match.

4.25 struct ServiceInterfaceTemplateRequest

An interface template.

Field	Type	Description
name	Name	Template name, snake_case.
protocol	String	Protocol of interfaces of this template.
propertyRequirements	List<ServiceInterfaceTemplateProperty>	Property rules of the template.

4.26 struct ServiceInterfaceTemplateResponse

A stored interface template.

Field	Type	Description
name	Name	Template name.
protocol	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
propertyRequirements	List<ServiceInterfaceTemplateProperty>	Property rules of the template.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.27 struct SystemListRequest

Systems to create or update.

Field	Type	Description
systems	List<SystemRequest>	Systems to create or update.

4.28 struct SystemListResponse

List of systems.

Field	Type	Description
entries	List<SystemResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.29 struct SystemQueryRequest

Paged filter for systems.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
systemNames	List<Name>	System names to match.
addresses	List<String>	Network addresses of the entity.
addressType	String	Address type filter: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
metadataRequirement List	List<MetadataRequirement>	Metadata requirements; each map is one alternative.
versions	List<Version>	System versions to match.
deviceNames	List<Name>	Device names to match.

4.30 struct SystemRequest

A system on behalf of which an operator registers.

Field	Type	Description
name	Name	System name, PascalCase.
metadata	Metadata	Free-form metadata (JSON object).
version	Version	Version of the system.
addresses	List<String>	Network addresses of the entity.
deviceName	Name	Name of the device hosting the system.

4.31 struct SystemResponse

A registered system.

Field	Type	Description
name	Name	System name.
metadata	Metadata	Free-form metadata (JSON object).
version	Version	Version of the system.
addresses	List<Address>	Network addresses with their types.
device	DeviceResponse	The device hosting the system, if registered.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.32 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD

document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

5 References

[1] J. Delsing, “ServiceRegistry – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, serviceregistry/documentation/v5.3.0/ServiceRegistry-v5.3.0-SysD.pdf.

[2] ——, “serviceRegistryManagement HTTP/TLS/JSON – Interface Design Description (ServiceRegistry),” Eclipse Arrowhead, IDD 5.3.0, 2026, serviceRegistry-v5.3.0-serviceRegistryManagement-HTTP-IDD.pdf.

[3] ——, “serviceRegistryManagement MQTT/TLS/JSON – Interface Design Description (ServiceRegistry),” Eclipse Arrowhead, IDD 5.3.0, 2026, serviceRegistry-v5.3.0-serviceRegistryManagement-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	